Structured questions

|!|BCELA0270000001|!| ★★ **STSE** **CRO**

Edexcel A Level-Biology B Paper 1 June 2017 Q7

Mitochondrial disorders may be caused by mutations in the genes coding for mitochondrial components. Some of these genes are found in mitochondrial DNA (mtDNA) and some are found in nuclear DNA.

Leigh syndrome is an example of a mitochondrial disorder. In this syndrome, a number of different proteins involved in respiration are affected.

These mutations may be inherited or may occur when DNA replicates.

(a) Explain why mutations in nuclear DNA can be inherited from either the mother or the father whereas mutations in mtDNA are only inherited from the mother. (2 marks)

(b) Some people with Leigh syndrome have a mutation in the MT-ATP6 gene. This gene codes for ATP synthase. This is a point mutation at nucleotide 8993 that changes thymine to guanine.

(i) What type of mutation is this? (1 mark)

- ☐ insertion
- ☐ monosomy
- ☐ substitution
- ☐ translocation

(ii) Explain how this mutation could affect oxidative phosphorylation. (3 marks)

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(c) Leigh syndrome can also be due to a mutation in the SURF1 gene. This mutation results in a shortened protein. Explain how a mutation can result in a shortened protein being produced. (2 marks)

(d) Some people with Leigh syndrome have mutations that affect proteins involved in the electron transport chain. Explain why these mutations lead to a build-up of lactate (lactic acid). (3 marks)

##

(a) Because nuclear DNA is present in both the male and female gametes, (1)
but mitochondria are present in the female gamete / ovum / egg cell and not the sperm head (which is the only component of a sperm that fuses with an ovum) (1)

(b) (i) ☑ substitution (1)

(ii) It will change the amino acid sequence / structure of ATP synthase (1)
therefore the active site of the ATP synthase may have a different structure / shape that ADP cannot bind (1)
therefore no or less ATP made / oxidative phosphorylation. (1)

(c) The mutation may produce a stop codon (1)
therefore translation will stop sooner / when the stop codon is reached. (1)
OR
One or several bases are deleted (1)
therefore the mRNA is shorter. (1)

(d) These mutations may prevent electrons being passed down the electron transport chain / ATPase from working properly (1)
therefore NADH cannot be oxidized (1)
therefore pyruvate and NADH will produce lactate (lactic acid) (1)
at a faster rate than it can be removed.

(11 marks)

##

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	Skills assessed	Marks
(a)	Applying concepts	2
(b)(i)	Understanding basic concepts	1
(b)(ii)	Applying concepts	3
(c)	Applying concepts	2
(d)	Applying concepts	3

|!|BCELA0270000002|!| ★★ **CRO**

Edexcel A Level-Biology B Paper 2 June 2017 Q3

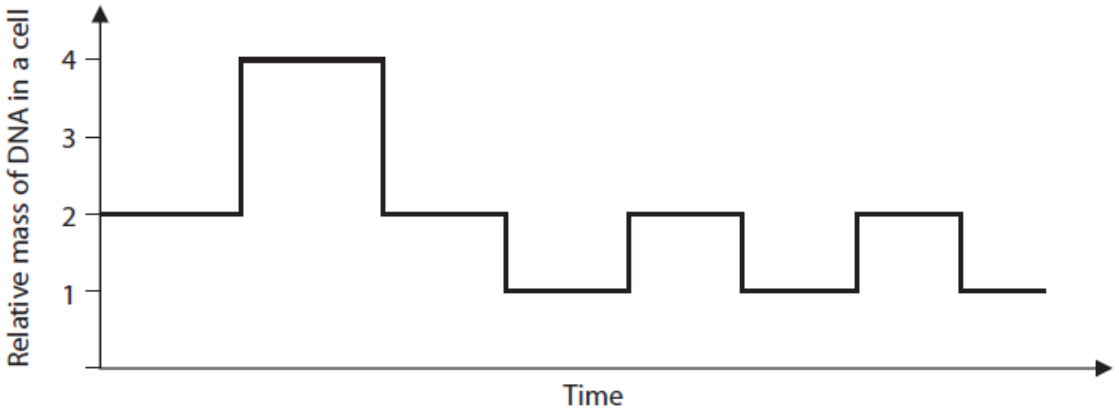
In flowering plants, pollen grains grow pollen tubes that are involved in fertilization, producing a zygote.

(a) (i) Which row of the table shows the correct description for these nuclei? (1 mark)

	Zygote nucleus	Pollen tube nucleus
☐	diploid	haploid
☐	haploid	haploid
☐	diploid	diploid
☐	haploid	diploid

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(ii) The production of plant gamete cells involves both mitosis and meiosis. The graph shows the DNA content of a cell during the production of a pollen grain. (1 mark)



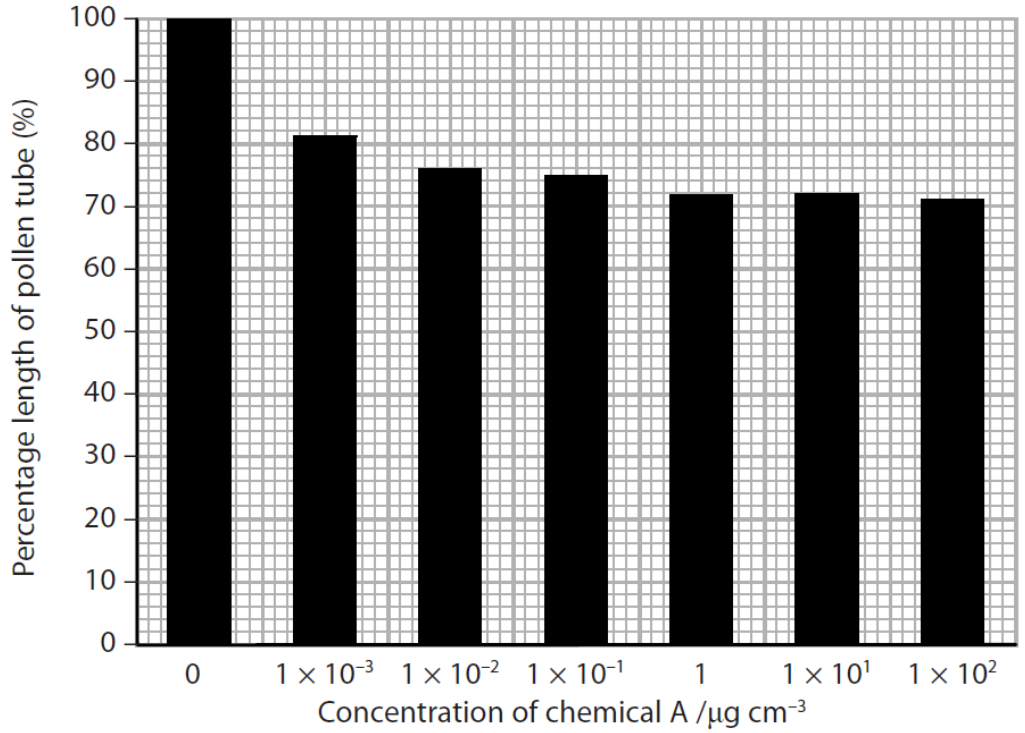
The order of nuclear divisions shown in the graph is

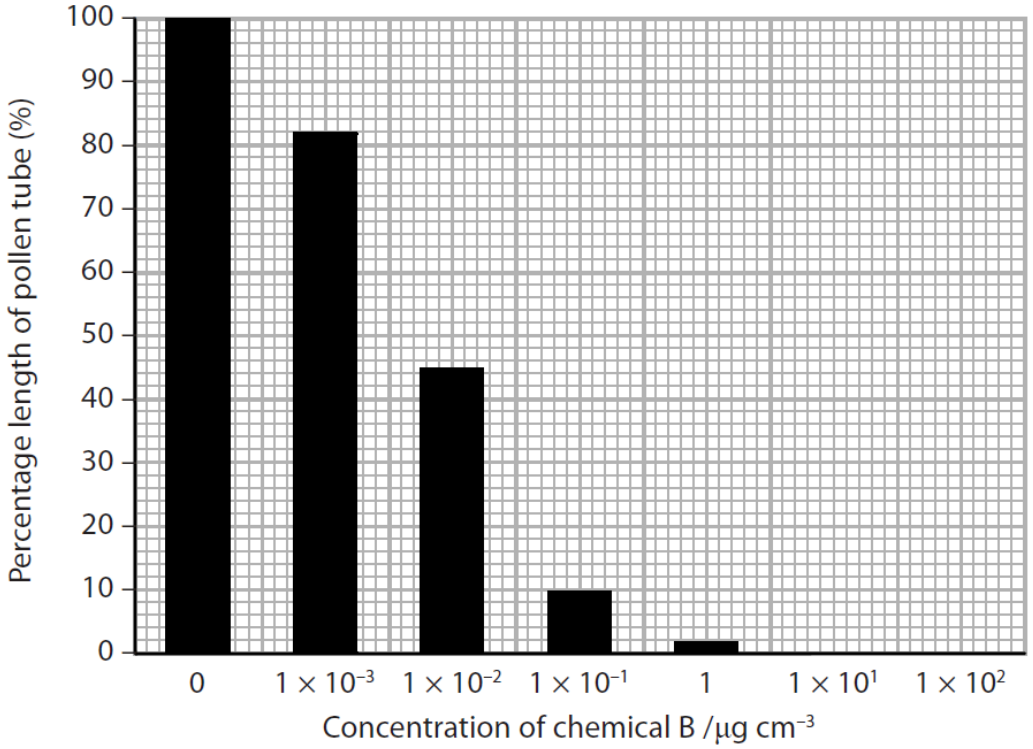
- ☐ meiosis, mitosis, mitosis
- ☐ meiosis, mitosis, meiosis
- ☐ mitosis, meiosis, meiosis
- ☐ mitosis, mitosis, meiosis

(b) A scientist investigated the effects of chemical A and chemical B on pollen tube growth. Chemical A prevents transcription. Chemical B prevents translation.

Pollen grains were treated with different concentrations of these two chemicals. The lengths of the pollen tubes they produced were compared with a control group of pollen grains. The lengths of the pollen tubes were measured and expressed as a percentage of the control group. The results are shown in the graphs.

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(i) The mean length of the pollen tubes in the control group was 7.5 mm. Calculate the mean length of the pollen tubes after adding 0.010 μg cm^{-3} of chemical A. (2 marks)

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(ii) Analyse the data to comment on the effects of chemical A and chemical B on the roles of transcription and translation in pollen tube growth. (4 marks)

##

(a) (i)

	Zygote nucleus	Pollen tube nucleus
☑	diploid	haploid

(1)

(ii) ☑ meiosis, mitosis, mitosis (1)

(b) (i) reading from graph at $1 \times 10^{-2} = 76\%$ (1)

$7.5 \times 76\% = 5.7$ mm (1)

(ii) Chemical A which prevents transcription has less effect / smaller reduction on pollen tube growth. (1)

The effect levels off when chemical A increases, but growth falls steeply / drops to zero when chemical B increases. (1)

mRNA is already present (1)

(pollen tube) growth requires production of new proteins / enzymes (1)

(8 marks)

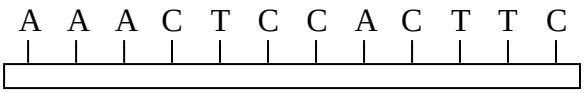
##

	Skills assessed	Marks
(a)(i)	Understanding basic concepts	1
(a)(ii)	Analysing data	1
(b)(i)	Mathematical skills	2
(b)(ii)	Analysing data	2
	Applying concepts	2

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|!|BCELA0270000003|!| ★★

The figure below shows part of the template strand of DNA.



(a) Give the corresponding mRNA sequence for the part of the template strand shown above. (2 marks)

The table below shows the mRNA codons for some amino acids and signals.

Amino acid / signal	**mRNA codons**					
Met (Start signal)	AUG					
Trp	UGG					
Phe	UUU	UUC				
Glu	GAA	GAG				
Lys	AAA	AAG				
Pro	CCU	CCC	CCA	CCG		
Val	GUU	GUC	GUA	GUG		
Gly	GGU	GGC	GGA	GGG		
Ser	UCU	UCC	UCA	UCG	AGU	AGC
Stop signal	UAA	UAG	UGA			

(b) Using the information from the table, give the amino acid sequence encoded by the mRNA sequence in (a). (1 mark)

(c) Describe the change in the mRNA and amino acid sequences if the following gene mutations occur in the DNA template strand.

(i) The second base (base A) is substituted with a base G. (2 marks)

(ii) The ninth base (base C) is substituted with a base A. (2 marks)

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(iii) The seventh base (base C) is deleted. (2 marks)

##

(a) UUUGAGGUGAAG (2)

Concept of mark award:
Follow the rules of complimentary base paring (1) and
mRNA has U instead of T (1)

(b) Phe – Glu – Val – Lys (1)

(c) (i) The first codon UUU will turn into UCU, (1)
which will code for Ser instead of Phe. (1)

(ii) The third codon GUG will turn into GUU, (1)
but the amino acid coded (Val) will not be changed. (1)

(iii) The third codon GUG will turn into UGA, which specifies a stop signal. (1)
The translation process will come to an end and the resulting polypeptide will have a shorter amino acid sequence. (1)

(9 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	2
(b)	Analysing data	1
(c)(i)	Analysing data	2
(c)(ii)	Analysing data	2
(c)(iii)	Analysing data	1
	Applying concepts	1

|!|BCELB0270000004|!| ★

The base sequence on a segment of mRNA and the amino acid sequence of the corresponding segment of polypeptide are shown below.

Base sequence on the mRNA	GAAGGUGCAUUCACCGGC
Amino acid sequence of the polypeptide	Glu – Gly – Ala – Phe – Thr – Gly

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(a) Name the process by which mRNA is formed. State the location in a cell where this process occurs. (2 marks)

(b) Give the sequence of bases on the coding strand of DNA which corresponds to the segment of mRNA shown above. (2 marks)

(c) Name the process by which the base sequence on mRNA is converted into the amino acid sequence of a polypeptide. State the organelle that is responsible for this process. (2 marks)

(d) Two different substitution mutations occur in the above segment of mRNA.
Mutation 1: the second base (base A) is substituted with a base C
Mutation 2: the last base (base C) is substituted with a base U
Which of the two mutations is more likely to result in a non-functional protein? Explain your answer. (5 marks)

##

(a) Transcription (1)
Nucleus (1)

(b) GAAGGTGCATTCACCGGC (2)

Concept of mark award:
The coding strand of DNA is the same as the mRNA (1)
except having T instead of U (1)

(c) Translation (1)
Ribosome (1)

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(d) Mutation 1 (1)

In mutation 1, the first codon GAA turns into GCA. Thus, the resulting polypeptide has one different amino acid, Alu instead of Glu. (1)

This change may affect the shape (conformation) and subsequently the function of the protein. (1)

In mutation 2, the last codon GGC turns into GGU, which still codes for the same amino acid Gly. (1)

The resulting polypeptide will be the same. (1)

(11 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Applying concepts	2
(c)	Understanding basic concepts	2
(d)	Analysing data	2
	Applying concepts	3

|!|BCELB0270000005|!| ★ **CRO**

The figures below show part of the karyotypes of two foetuses, P and Q, with different chromosomal abnormalities.

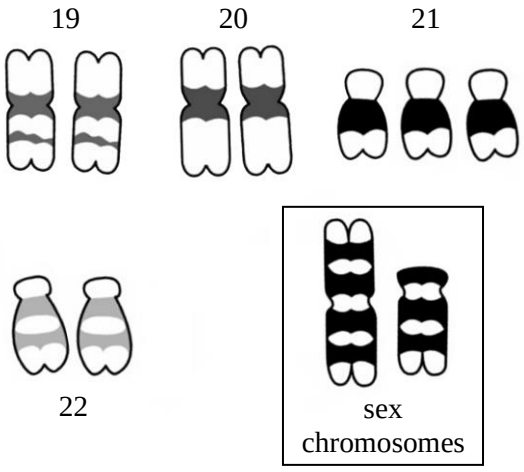


Figure 1 Chromosomes 19–22 and sex chromosomes of P

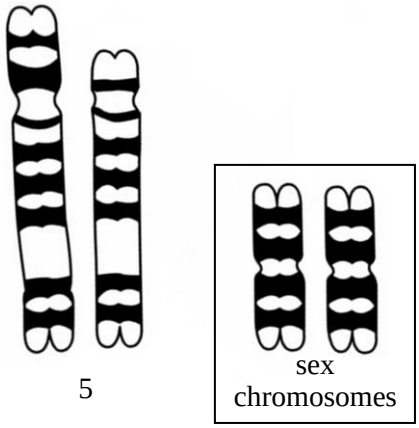


Figure 2 Chromosomes 5 and sex chromosomes of Q

(a) What is the sex of P and Q? Explain your answer. (4 marks)

(b) (i) What genetic disorder does foetus P probably have? (1 mark)

(ii) Suggest how the chromosomal abnormality shown in Figure 1 may arise. (3 marks)

(c) State the chromosomal abnormality shown in Figure 2. Suggest ***one*** cause for this chromosomal abnormality. (2 marks)

##

(a) P is a male. (1)
The two sex chromosomes are different in size/length and thus one is X chromosome and the other one is Y chromosome. (1)
Q is a female. (1)
The two sex chromosomes are of the same size/length and thus both are X chromosomes. (1)

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(b) (i) Down syndrome (1)

(ii) During gamete formation / meiotic cell division, the two homologous members / sister chromatids of chromosome 21 fail to separate. (1)

Therefore, some of the gametes formed contain two copies of chromosome 21. (1)

If a gamete containing an extra copy of chromosome 21 fuses with a normal gamete during fertilization, the resulting zygote will have three copies of chromosome 21. (1)

(c) One copy of chromosome 5 is shorter/longer than the other. (1)

Any ***one*** of the following causes: (1)

- Deletion / duplication of a segment of chromosome 5
- A segment of another chromosome becomes attached to chromosome 5

(Accept other reasonable answers.)

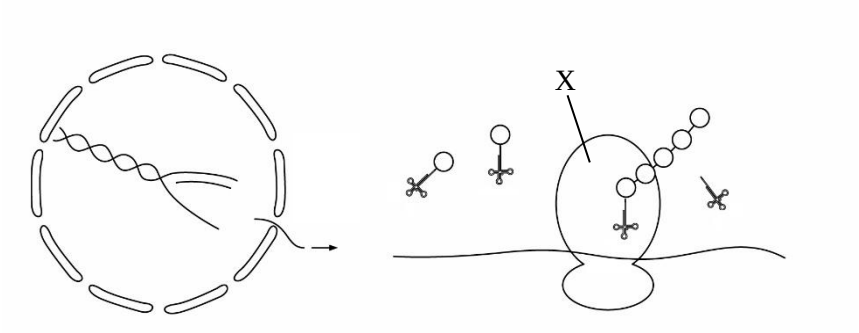
(10 marks)

##

	Skills assessed	Marks
(a)	Analysing data	4
(b)(i)	Analysing data	1
(b)(ii)	Applying concepts	3
(c)	Analysing data	1
	Applying concepts	1

|!|BCELA0270000006|!| ★★ **CRO**

Insulin is a protein hormone that helps regulate blood glucose levels. The diagram below shows two cellular processes involved in the production of insulin.



Process A Process B

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(a) Identify processes A and B. (2 marks)

(b) (i) What is structure X? (1 mark)

(ii) Describe the role of structure X in protein synthesis. (2 marks)

(c) An insulin molecule consists of 51 amino acids. What is the minimum number of nucleotide base pairs that you would expect in the gene that codes for insulin? (1 mark)

(d) The base sequence on a segment of the template strand of DNA in the gene for insulin is shown below.

CCATAGCACCTCGTTACA

Give the base sequence of the mRNA complementary to this segment of DNA. (1 mark)

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(e) The table below shows the 64 mRNA codons and the corresponding amino acids and signals they specify.

First base	Second base				Third base
	U	**C**	**A**	**G**	
U	Phe	Ser	Tyr	Cys	**U**
	Phe	Ser	Tyr	Cys	**C**
	Leu	Ser	(Stop)	(Stop)	**A**
	Leu	Ser	(Stop)	Trp	**G**
C	Leu	Pro	His	Arg	**U**
	Leu	Pro	His	Arg	**C**
	Leu	Pro	Gln	Arg	**A**
	Leu	Pro	Gln	Arg	**G**
A	Ile	Thr	Asn	Ser	**U**
	Ile	Thr	Asn	Ser	**C**
	Ile	Thr	Lys	Arg	**A**
	Met (Start)	Thr	Lys	Arg	**G**
G	Val	Ala	Asp	Gly	**U**
	Val	Ala	Asp	Gly	**C**
	Val	Ala	Glu	Gly	**A**
	Val	Ala	Glu	Gly	**G**

Using the information from the table, give the amino acid sequence encoded by the mRNA sequence in (d). (1 mark)

_____

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(f) In some people, the segment of DNA was found showing a different sequence of bases caused by a mutation:

CCATAGCACATCGTTACA

(i) What type of mutation has occurred to this sequence? (1 mark)

(ii) Explain the phenotype of this mutation that you would expect regarding the regulation of blood glucose. (3 marks)

##

(a) Process A: transcription (1)
Process B: translation (1)

(b) (i) Ribosome (1)

(ii) Ribosomes facilitate the pairing of tRNA anticodons with mRNA codons and the formation of peptide bonds between amino acids to build a polypeptide. (1) (1)

(c) $51 \times 3 = 153$ (1)

(d) GGUAUCGUGGAGCAAUGU (1)

(e) Gly – Ile – Val – Glu – Gln – Cys (1)

(f) (i) Substitution (1)

(ii) The fourth codon on the mRNA transcribed will become UAG, which is a stop codon. (1)
The production of the polypeptide will be stopped prematurely. (1)
Without insulin, the body cannot regulate the amount of glucose in blood resulting in high blood glucose levels / diabetes. (1)

(12 marks)

##

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	Skills assessed	Marks
(a)	Analysing data	2
(b)(i)	Analysing data	1
(b)(ii)	Understanding basic concepts	2
(c)	Mathematical skills	1
(d)	Applying concepts	1
(e)	Analysing data	1
(f)(i)	Analysing data	1
(f)(ii)	Analysing data	1
	Applying concepts	2

|!|BCELB0270000007|!| ★

(a) Describe how an mRNA molecule is transcribed from DNA. (4 marks)

(b) Give ***two*** structural differences between mRNA and tRNA. (2 marks)

(c) Analysis of a particular polypeptide shows that the gene involved in its synthesis is 330 base pairs long yet the polypeptide has only 84 amino acids.

(i) What is the minimum number of base pairs required to code for 84 amino acids? Explain your answer. (2 marks)

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(ii) Suggest why the gene contains 330 base pairs. (1 mark)

##

(a) The hydrogen bonds that hold the two DNA strands of a gene break. / The double helix unwinds. (1)
Free RNA nucleotides pair with the exposed bases in the DNA template strand following the rules of complementary base pairing. (1)
RNA polymerase catalyzes the joining of the adjacent RNA nucleotides to form an mRNA strand. (1)

(b) Any ***two*** of the following: (2)
- mRNA has no base-pairing while tRNA has base-pairing.
- mRNA is linear/straight, while tRNA is cloverleaf-shaped.
- mRNA has no binding site for amino acids, while tRNA has.

(c) (i) 252 (or 255) (1)
Every three bases form a codon which codes for one amino acid (and a stop codon terminates the translation process). (1)

(ii) The gene contains non-coding sections. / Part of the polypeptide is subsequently removed. (1)

(9 marks)

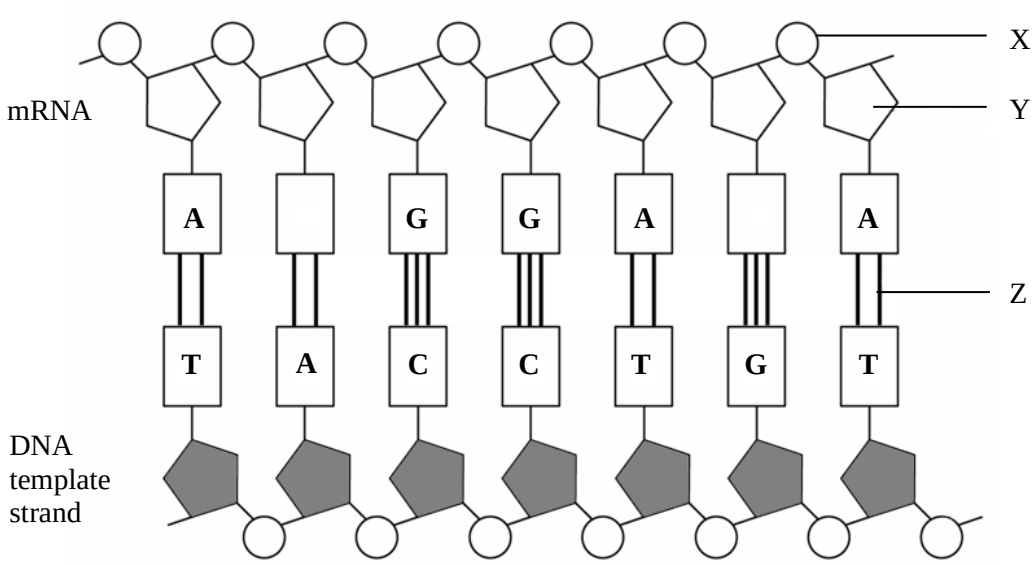
##

	Skills assessed	Marks
(a)	Understanding basic concepts	4
(b)	Understanding basic concepts	2
(c)(i)	Mathematical skills	1
	Applying concepts	1
(c)(ii)	Applying concepts	1

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|!|BCELC0270000008|!|

The diagram below shows a segment of a DNA template strand and the mRNA produced from it during transcription.



(a) What do X, Y and Z represent? (3 marks)

(b) Complete the diagram to show the sequence of bases on the mRNA. (1 mark)

(c) Starting with mRNA, describe how the process of translation leads to the production of a polypeptide. (5 marks)

##

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(a) X: phosphate group (1)
Y: ribose (1)
Z: hydrogen bond (1)

(b) U (pairs up with A) and C (pairs up with G) (1)

(c) The mRNA produced in transcription leaves the nucleus and enters the cytoplasm. (1)
The mRNA attaches to a ribosome. (1)
Each tRNA molecule with a particular anticodon only carries a specific amino acid. (1)
tRNA with anticodons complementary to the codons of the mRNA will attach to the ribosome in sequence. (1)
Adjacent amino acids are joined by peptide bonds to form a polypeptide. (1)

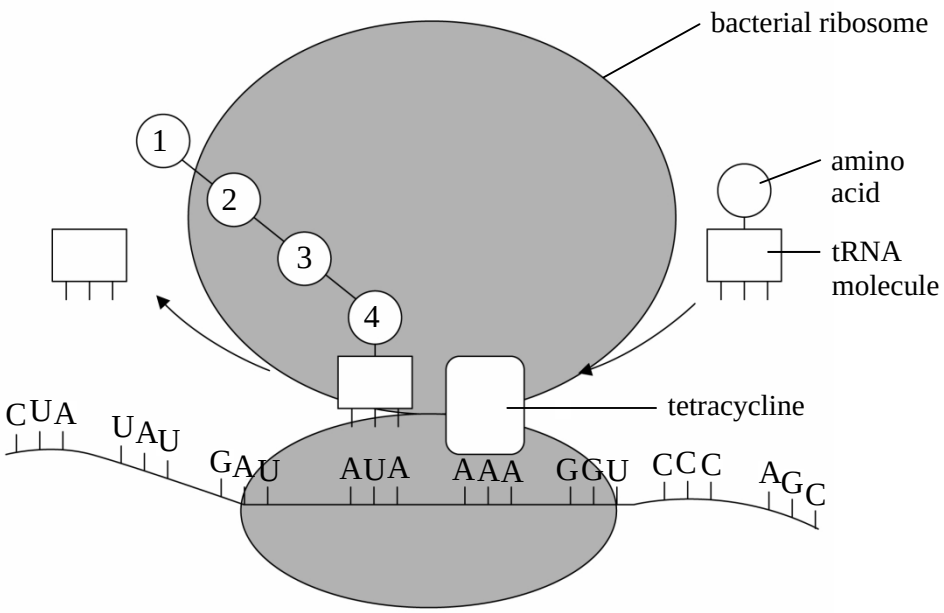
(9 marks)

##

	Skills assessed	Marks
(a)	Analysing data	3
(b)	Applying concepts	1
(c)	Understanding basic concepts	5

|!|BCELA0270000009|!| ★★ **CRO**

Protein synthesis in bacteria is similar to that in eukaryotic cells. The diagram below shows how the antibiotic tetracycline interferes with an event in protein synthesis in a bacterial cell.



(a) Identify the stage of protein synthesis shown in the diagram. (1 mark)

(b) The table below shows the anticodons of certain tRNA molecules and the corresponding amino acids that these molecules carry.

tRNA anticodon	Amino acid
AUA	Ile
CCA	Pro
CUA	Leu
GAU	Asp
UAU	Thr
UUU	Phe

Give the sequence of the four amino acids in the polypeptide shown in the diagram. (1 mark)

(c) Give the base sequence on the coding strand of DNA coding for the amino acid sequence in (b). (1 mark)

(d) Tetracycline prevents bacterial growth by preventing protein synthesis.

(i) With reference to the diagram, explain how tetracycline stops protein synthesis in bacteria. (2 marks)

(ii) Give ***two*** other ways in which antibiotics can prevent bacterial growth. (2 marks)

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##

(a) Translation (1)

(b) Asp – Ile – Leu – Thr (1)

(c) CTATATGATATA (1)

(d) (i) Tetracycline blocks a site on the bacterial ribosome where a tRNA molecule binds to an mRNA codon. (1)

This prevents tRNA from bringing amino acids to the ribosomes for building the polypeptide. (1)

(ii) Any ***two*** of the following: (2)

- Disrupts cell walls / prevents cell wall synthesis
- Disrupts cell membranes / prevents synthesis of cell membranes
- Prevents DNA replication / synthesis of nucleic acids

(7 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Analysing data	1
(c)	Applying concepts	1
(d)(i)	Applying concepts	2
(d)(ii)	Understanding basic concepts	2

|!|BCELB0270000010|!| ★

(a) The base sequence of a section of mRNA is shown below.

UACAGAGCAUCGUUA

(i) Give the base sequence of the DNA template from which the mRNA is transcribed. (1 mark)

(ii) Name the enzyme involved in the transcription of mRNA. (1 mark)

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(iii) The table below shows the 64 mRNA codons and the corresponding amino acids and signals they specify.

First base	Second base				Third base
	U	C	A	G	
U	Phe	Ser	Tyr	Cys	**U**
	Phe	Ser	Tyr	Cys	**C**
	Leu	Ser	(Stop)	(Stop)	**A**
	Leu	Ser	(Stop)	Trp	**G**
C	Leu	Pro	His	Arg	**U**
	Leu	Pro	His	Arg	**C**
	Leu	Pro	Gln	Arg	**A**
	Leu	Pro	Gln	Arg	**G**
A	Ile	Thr	Asn	Ser	**U**
	Ile	Thr	Asn	Ser	**C**
	Ile	Thr	Lys	Arg	**A**
	Met (Start)	Thr	Lys	Arg	**G**
G	Val	Ala	Asp	Gly	**U**
	Val	Ala	Asp	Gly	**C**
	Val	Ala	Glu	Gly	**A**
	Val	Ala	Glu	Gly	**G**

Using the information from the table, determine the amino acid sequence encoded by the mRNA sequence. (1 mark)

(iv) Suggest how the cell ensures that the correct amino acids are added to the polypeptide. (2 marks)

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(b) Proflavin is a chemical that can change the base sequence of DNA.

(i) What is the name given to this type of chemical? (1 mark)

(ii) If proflavin causes the deletion of the first base in the DNA sequence from which the above mRNA is transcribed, explain how this leads to the production of a non-functional protein. (3 marks)

##

(a) (i) ATGTCTCGTAGCAAT (1)

(ii) RNA polymerase (1)

(iii) Tyr – Arg – Ala – Ser – Leu (1)

(iv) Each type of tRNA carries a specific type of amino acid. (1)
The anticodon of a given tRNA only binds to its complimentary mRNA codons. (1)

(b) (i) Mutagen (1)

(ii) The deletion of the base will shift the reading frame. (1)
Thus, the whole amino acid sequence after the point of deletion is altered. (1)
This affects the shape (conformation) and subsequently the function of the protein. (1)

(9 marks)

##

	Skills assessed	Marks
(a)(i)	Applying concepts	1
(a)(ii)	Understanding basic concepts	1
(a)(iii)	Analysing data	1
(a)(iv)	Applying concepts	2
(b)(i)	Understanding basic concepts	1
(b)(ii)	Applying concepts	3

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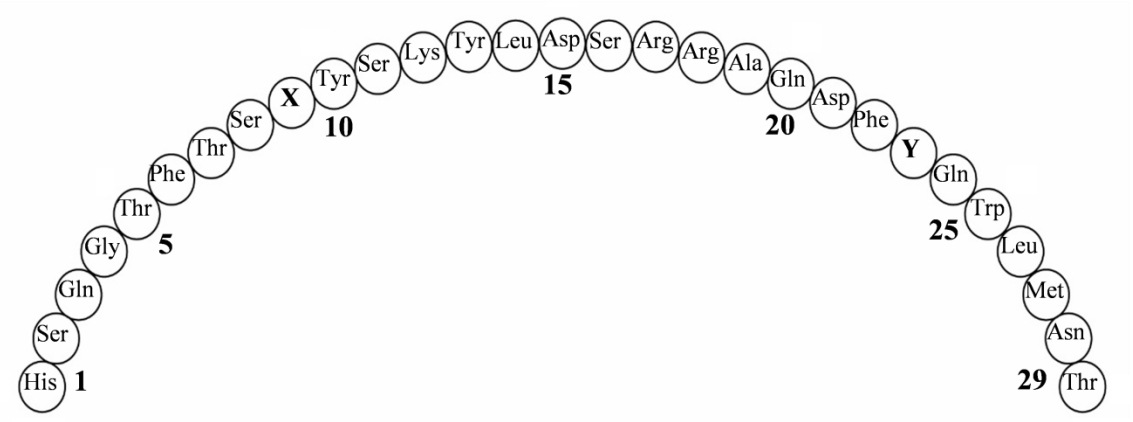
|!|BCELA0270000011|!| ★★ **CRO**

Glucagon is a protein hormone that is involved in the regulation of blood glucose.

(a) Which organ does glucagon act on? (1 mark)

(b) Explain how high levels of glucagon would affect the blood glucose level. (2 marks)

The diagram below shows the sequence of amino acids in the polypeptide of glucagon. (The numbers represent the positions of amino acids in the polypeptide.)



The base sequence on the template strand of DNA corresponding to the above polypeptide is shown below. (The numbers represent the position of the bases in the DNA strand.)

GTG	AGC	GTC	CCG	TGT	AAG	TGG	TCA[24]
CTG	ATG	TCG	TTC	ATG	GAC	CTG	AGG[48]
TCC	GCA	CGG	GTC	CTG	AAG	CAC	GTC[72]
ACC	AAC	TAC	TTA	TGG[87]			

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The table below shows the 64 mRNA codons and the corresponding amino acids and signals they specify.

First base	Second base				Third base
	U	**C**	**A**	**G**	
U	Phe	Ser	Tyr	Cys	**U**
	Phe	Ser	Tyr	Cys	**C**
	Leu	Ser	(Stop)	(Stop)	**A**
	Leu	Ser	(Stop)	Trp	**G**
C	Leu	Pro	His	Arg	**U**
	Leu	Pro	His	Arg	**C**
	Leu	Pro	Gln	Arg	**A**
	Leu	Pro	Gln	Arg	**G**
A	Ile	Thr	Asn	Ser	**U**
	Ile	Thr	Asn	Ser	**C**
	Ile	Thr	Lys	Arg	**A**
	Met (Start)	Thr	Lys	Arg	**G**
G	Val	Ala	Asp	Gly	**U**
	Val	Ala	Asp	Gly	**C**
	Val	Ala	Glu	Gly	**A**
	Val	Ala	Glu	Gly	**G**

(c) Identify amino acids X and Y shown in the diagram. (2 marks)

_____

_____

(d) Give an amino acid in the polypeptide that corresponds to only one type of tRNA. Give the anticodon of the tRNA. (2 marks)

_____

_____

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(e) Two different mutations occur in the above segment of DNA.
Mutation 1: the 26th base (base T) is substituted with a base C
Mutation 2: the 69th base (base C) is substituted with a base T
Which of the two mutations is more likely to result in the production of non-functional glucagon? Explain your answer. (5 marks)

##

(a) Liver (1)

(b) High levels of glucagon stimulate the conversion of glycogen to glucose in the liver. (1)
The glucose is then released into the blood, increasing the blood glucose level. (1)

(c) Amino acid X: Asp (1)
Amino acid Y: Val (1)

(d) Try / Met (1)
ACC / UAC (1)

(e) Mutation 1 (1)
In mutation 1, the codon for the 9th amino acid changes from GAC to GGC. (1)
Thus, the resulting polypeptide has one different amino acid, Gly instead of Asp.
The polypeptide may fold into a different shape and become non-functional. (1)
In mutation 2, the codon for the 23th amino acid changes from GUG to GUA (1)
which still codes for the same amino acid Gly. (1)

(12 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	2
(c)	Analysing data	2
(d)	Analysing data	2
(e)	Analysing data	4
	Applying concepts	1

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